ASSIGNMENT-8

①

x	0.5	1.0	1.5	2.0
$f(x)$	0.9197	0.6767	0.3734	0.1831

$f'(x)$ @ $x=1$

$x_i = 1, \quad h = 0.5$

$f(x_{i-1}) = f(0.5) = 0.9197$

$f(x_i) = f(1) = 0.6767$

$f(x_{i+1}) = f(1.5) = 0.3734$

$f(x_{i+2}) = f(2) = 0.1831$

(a) $f'(x_i) = \dfrac{f(x_{i+1}) - f(x_i)}{h}$

$f'(1.0) = \dfrac{0.3734 - 0.6767}{0.5} = -0.6066$

(b) $$f'(x_i) = \frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h}$$

$$f'(1) = \frac{-0.1831 + 4(0.3734) - 3(0.6767)}{2(0.5)}$$

$$f'(1) = \frac{-0.1831 + 1.4936 - 2.0301}{} = -0.7196$$

(c) $$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} \simeq \frac{0.6767 - 0.9197}{0.5}$$

$$f'(1) = \frac{-0.2430}{0.5} = -0.4860$$

(d) $$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h}$$

$$f'(1) = \frac{0.3734 - 0.9197}{2(0.5)}$$

$$f'(1) = -0.5463/1 = -0.5462$$

(c) $f(x) = 5xe^{-2x}$

$$f'(x) = 5\left(e^{-2x} + x(-2)e^{-2x}\right)$$

$$= 5e^{-2x}(1-2x)$$

$$f'(1) = 5e^{-2}(-1) = -5e^{-2} \simeq -0.6767$$

True errors:

(i) error in (a): $-0.6767 - (-0.6066)$

$$= |-0.0701| = 0.0701$$

(ii) error in (b) $= -0.6767 - (-0.7196)$

$$= |0.0429| = 0.0429$$

(iii) error in (c) $= -0.6767 - (-0.4860)$

$$= |-0.1907| = 0.1907$$

(iv) error in (d) $= -0.6767 - (-0.5463)$

$$= |-0.1304| = 0.1304.$$

→ Most accurate estimate

↳(b)→ Forward divided difference

$O(h^2)$

Exp :- Say we take $f''(x)$

$$f(x) = 5xe^{-2x}$$

$$f'(x) = 5e^{-2x}(1-2x)$$

$$f''(x) = 5\left(e^{-2x}(-2)(1-2x) + e^{-2x}(-2)\right)$$

$$= 5e^{-2x}(-2+4x-2)$$

$$= 5e^{-2x}(4x-4) = 20e^{-2x}(x-1)$$

$$= 20e^{-2x}(x-1)$$

$f''(1) = 0$

↳ This implies the derivative $f(x)$ reaches a local min

@ $x = 1$.

$\therefore$ $x = 1$ is the lowest point of the curve $f'(x)$, slopes calculated @ the $x = 0.5$ & $x = 1.5$ are both significantly less −ve ⟶ 0 than at the min.

$\therefore$ The central diff. method (d) which averages the trend across $x = 0.5$ to $x = 1.5$, → It grossly under estimates the Steepness of the min

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Leads to poor Approximation even though it is $O[h^2]$

→ But in (b) otherway s.

it fits a Quadratic curve exclusively using the points to the right of the min so, missing it allows it to adapt more accurately to the upward swing of the derivative's curve, without spanning across the ~~state~~ bottom of the minimum.

(5) $h=1$

(a)

$$RQ(t) = \frac{R_{CO_2}(t)}{R_{O_2}(t)} = \frac{\frac{d(CO_2)}{dt}}{\frac{d(O_2)}{dt}}$$

① forward difference $O(h^1)$ @ $t=0$

$$f'(x_0) = \frac{-f(x_2) + 4f(x_1) - 3f(x_0)}{2h}$$

$$R_{CO_2}(0) = \frac{-151 + 4(75) - 3(0)}{2(1)} = \frac{149}{2} = 74.5$$

$$R_{O_2}(0) = \frac{-149 + 4(76) - 3(0)}{2(1)} = \frac{155}{2} = 77.5$$

$$RQ(0) = \frac{74.5}{77.5} = 0.9613$$

2. Central divided difference:

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h}$$

@ t = 1;

$R_{CO_2}(1) = \frac{151}{2} = 75.5$

$R_{O_2}(1) = \frac{149}{2} = 74.5$

$\frac{75.5}{74.5} = \boxed{1.0134}$ → RQ(1)

@ t = 2:

$R_{CO_2}(2) = \frac{225 - 75}{2} = 75$

$R_{O_2}(2) = \frac{224 - 76}{2} = 74$

$RQ(2) = \frac{75}{74} = 1.0135$

@ t = 3

$R_{CO_2}(3) = \frac{297 - 151}{2} = 73.$

$R_{O_2}(3) = \frac{301 - 149}{2} = 76$

$RQ(3) = \frac{73}{76} = 0.9605$

@ t = 4,

$$R_{CO_2}(4) = \frac{367 - 225}{2} = 71$$

$$R_{O_2}(4) = \frac{381 - 229}{2} = 78.5$$

$$RQ(4) = \frac{71}{78.5} = 0.9045$$

@ t = 5

$$R_{CO_2}(5) = \frac{436 - 297}{2} = 69.5$$

$$R_{O_2}(5) = \frac{462 - 300}{2} = 81$$

$$RQ(5) = \frac{69.5}{81} = 0.8580$$

and we continue soon

t	R_{CO_2}	R_{O_2}	RQ
6	67.5	82	0.8133
7	66	84	0.7857
8	66	84	0.7857
9	66.5	87	0.7824

@ t = 10; Backward difference $O(h^2)$:

$$f'(x_0) = \frac{3f(x_{10}) - f(x_9)4 + f(x_8)}{2h}$$

$$R_{CO_2}(10) = \frac{3(701) - 4(634) + 568}{2(1)}$$

$$= \frac{2103 - 2536 + 568}{2} = 67.5$$

$$R_{O_2}(10) = \frac{3(801) - 4(715) + 681}{1} = 87$$

RQ(10) = 67.5/87 = 0.7759.

(5) Total :- $\frac{h}{2}[f(x_0) + 2(f(x_1) + \cdots f(x_9)) + f(x_{10})]$

$\frac{1}{2}[74.5 + 2(75.5 + 75 + 73 + 71 + 69.5 + 69.5 + 66 + 66 + 66.5) + 67.5)$

Total CO_2 = $\frac{1}{2}(74.5 + 1260 + 67.5)$

Total CO_2 ⇒ 701 g

(6) Total : $\frac{h}{3}[f(x_0) + 4\Sigma f(x_{odd}) + 2(\Sigma f(x_{even})) + f(x_{10})]$

Total CO_2 =

$\frac{1}{3}[74.5 + 4(R_{CO_2}(1) + R_{CO_2}(3) + R_{CO_2}(5) + R_{CO_2}(7) + R_{CO_2}(9)) + 2(R_{CO_2}(2) + R_{CO_2}(4) + R_{CO_2}(6) + R_{CO_2}(8)) + 675]$

$$\Rightarrow \frac{1}{3}\left(34.5+4(75.5+73+69.5+66+66.5)\right.$$
$$\left.+2(75+71+63.5+66)\right.$$
$$\left.+67.5\right)$$

$$\text{Total} = \frac{1}{3}[34.5+1402+559+67.5]$$

$$\text{Total} = 701.8$$

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